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Set-A - Q1 (ECL)

(a) $V_X = V_Y = \text{High}$.

$\therefore Q_1, Q_2$ in active mode, Q_R cutoff

$$\therefore V_{C2} = 0V$$

$$\therefore \boxed{V_{OR}} = V_{C2} - 0.7 = -0.7V \quad \boxed{\text{Ans.}}$$

$$\therefore V_X = V_Y = \text{High} = -0.7V$$

$$\therefore V_E = V_X - 0.7 = -1.4V$$

$$\therefore I_E = \frac{-1.4 + 4.1}{0.9 \text{ k}\Omega} = 3 \text{ mA}$$

$$\therefore I_{C1} = I_E = 3 \text{ mA}$$

$$\begin{aligned} \therefore V_{C1} &= 0 - I_{C1}(R_{C1}) = 0 - (3)(0.368) \\ &= -1.104 \end{aligned}$$

$$\therefore \boxed{V_{NOR}} = V_{C1} - 0.7 = \boxed{-1.804} \quad \boxed{\text{Ans.}}$$

(b) $V_R = \frac{V_{\text{High}} + V_{\text{Low}}}{2} = \frac{-0.7 - 1.804}{2} = \boxed{-1.252V}$

$$\therefore V_X = V_R - 0.7 = \boxed{-1.952V}$$

$$I_1 = \frac{V_{NOR} + 4.1}{1.7 \text{ k}\Omega} = \frac{4.1 - 1.804}{1.7 \text{ k}\Omega} = \boxed{1.351 \text{ mA}}$$

$$I_2 = \frac{V_{OR} + 4.1}{1.7 \text{ k}\Omega} = \frac{4.1 - 0.7}{1.7 \text{ k}\Omega} = \boxed{2 \text{ mA}}$$

$$I_4 = \frac{V_X + 4.1}{1.2 \text{ k}\Omega} = \boxed{\cancel{1.2635 \text{ mA}}} \quad 1.79 \text{ mA}$$

Set-B (Q4) (ECL)

(a) V_x V_y High

$\therefore Q_1, Q_2$ in active mode, Q_R cutoff.

$$\therefore V_{OR} = V_{C2} - 0.7 = -0.7 \text{ V} \quad \boxed{\text{Ans.}}$$

$$\therefore V_x = V_y = \text{High} = -0.7 \text{ V}$$

$$\therefore I_E = \frac{-1.4 + 3.9}{0.75 \text{ k}\Omega} = 3.333 \text{ mA}$$

$$\begin{aligned} \therefore V_{C1} &= 0 - I_E(R_{C1}) = 0 - (3.333)(0.4) \\ &= -1.333 \text{ V} \end{aligned}$$

$$\therefore \boxed{V_{NOR}} = V_{C1} - 0.7 = -2.033 \text{ V} \quad \boxed{\text{Ans.}}$$

$$\textcircled{b} \quad \boxed{V_R} = \frac{V_{Hi} + V_{Low}}{2} = \frac{-0.7 - 2.033}{2} = \boxed{-1.367 \text{ V}}$$

$$\boxed{V_x} = V_R - 0.7 = \boxed{-2.067 \text{ V}}$$

$$\boxed{I_1} = \frac{-2.033 + 3.9}{1.7 \text{ k}\Omega} = \boxed{1.098 \text{ mA}}$$

$$\boxed{I_2} = \frac{-0.7 + 3.9}{1.7 \text{ k}\Omega} = \boxed{1.882 \text{ mA}}$$

$$\boxed{I_4} = \frac{-2.067 + 3.9}{1.343 \text{ k}\Omega} = \boxed{1.365 \text{ mA}}$$

Question No. 2

The circuit in Figure 2 has the following parameters: $\beta_F = 25$, $\beta_R = 0.1$, $V_{BE}(sat) = 0.8V$, $V_{BE}(active) = 0.7V$, $V_{CE}(sat) = 0.1V$ and $V_{OH} = 3.4V$

(a)	CO1	Briefly explain the advantages of using the totem-pole output stage.	[4]
(b)	CO2	Assume, $v_x = v_y = 5V$ and no loads are connected. Find all the currents in mA and voltages in V labeled in the Driver circuit of Figure 2 (including v_o).	[10]
(c)	CO2	Assume, $v_x = v_y = 0.1V$. Determine I_B , I_E , I_C and estimate the maximum number of fanout you can connect to the driver circuit in this case.	[6]

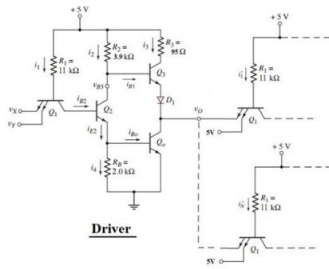


Figure 2

Set A: Q2
Set B: Q6

(a) Advantage:

- 1) Low pullup resistance.
- 2) charging time and discharging time almost same.

(b) If $V_x = V_y = 5V$, $Q_1 \rightarrow R.A$

$Q_2 \rightarrow$ Saturation, $Q_3 \rightarrow$ Saturation.

$Q_3 \rightarrow$ cutoff.

①

①

$$V_{BE} = 0.7V, V_{BE} = 0.8V, V_{BE} = 0.9V$$

$$I_1 = \frac{5 - 2.3}{11k\Omega} = 0.2454 \text{ mA} \quad ①$$

$$I_{B2} = I_1 + 2 \times 0.1 \times I_1 = 0.2958 \text{ mA} \quad ①$$

$$I_2 = \frac{5 - 0.9}{3.9k\Omega} = 1.0513 \text{ mA} \quad ①$$

$$I_B = 0 \text{ mA}, I_{B3} = 0 \text{ mA} \quad ①$$

$$I_{E2} = I_{B2} + I_2 = 1.3471 \text{ mA} \quad ①$$

$$I_{C1} = \frac{0.8 - 0}{2k\Omega} = 0.4 \text{ mA} \quad ①$$

$$I_{B3} = I_{E2} - I_{C1} = 0.9471 \text{ mA} \quad ①$$

(c) $V_x = V_y = 0.1V$,

So, $Q_1 \rightarrow$ saturation, $Q_2 \rightarrow$ cutoff, $Q_3 \rightarrow$ cutoff

$Q_3 \rightarrow$ active

Now, $V_{BE} = V_{OH} = 3.4V$.

$$V_{BE} = (V_{BE} + 2 \times 0.7)V = 4.8V$$

$$I_2 = I_{B3} = \frac{5 - 4.8}{3.9k\Omega} = 0.0513 \text{ mA}$$

$$\text{So, supply} = I_{E3} = (\beta + 1) \times I_2 = 26 \times I_2 = 1.33 \text{ mA} \quad ②$$

Demand

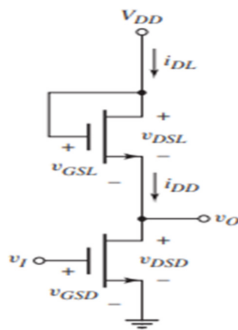
$$I_L = 0.1 \times 0.2454 = 0.02454 \text{ mA} \quad ②$$

$$\text{So, } N = \left\lfloor \frac{1.33}{0.02454} \right\rfloor = \lfloor 54.33 \rfloor = 54 \quad ②$$

SETA Ques 03

The enhancement-load NMOS inverter in the figure is biased at $V_{DD} = 3\text{ V}$. The transistor parameters are $K'_n = 100\mu\text{A/V}^2$ (use $K'_n \cdot (W/L)$ to calculate K_n), $V_{TND} = V_{TNL} = 0.4\text{ V}$, $(W/L)_D = 16$, and $(W/L)_L = 2$.

(a)	CO1	Identify the operating mode of the load NMOS transistor.	[4]
(b)	CO1	Determine the value of v_o in V and I_{DL} , I_{DD} in mA for $v_I = 0.1\text{ V}$ and $v_I = 1\text{ V}$	[8]
(c)	CO2	Design a static CMOS logic circuit that implements the logic function $Y = (A+B)+CD$. (indicate both PMOS and NMOS networks)	[4+4]



a) operating mode of Load NMOS,

$$v_{GSL} = v_{DSL}$$

$$\text{as } v_{DSL} > v_{DSL} - V_{TNL}$$

① saturation mode [4 marks]

b) $v_I = 0.1\text{ V}$ case I

$$\text{As } v_I < V_{TND} \rightarrow I_D = 0$$

$$I_D = K_{NL}(v_{DSL} - V_{TNL})^2 = 0$$

$$\Rightarrow v_{DD} - v_o - V_{TNL} = 0$$

$$\Rightarrow v_o = v_{DD} - V_{TNL} = (3 - 0.4)\text{ V} = 2.6\text{ V}$$

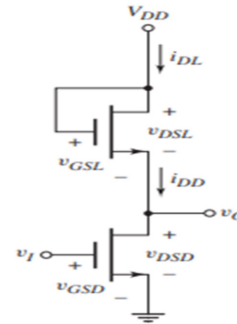
① $v_o = 2.6\text{ V}$ [2 marks]

②, ① $I_{DD} = I_{DL} = 0$ [1+1 mark]

SETB Ques 05

The enhancement-load NMOS inverter in the figure is biased at $V_{DD} = 3\text{ V}$. The transistor parameters are $K'_n = 90\mu\text{A/V}^2$ (use $K'_n \cdot (W/L)$ to calculate K_n), $V_{TND} = V_{TNL} = 0.5\text{ V}$, $(W/L)_D = 16$, and $(W/L)_L = 2$.

(a)	CO1	Identify the operating mode of the load NMOS transistor.	[4]
(b)	CO1	Determine the value of v_o in V and I_{DL} , I_{DD} in mA for $v_I = 0.2\text{ V}$ and $v_I = 0.9\text{ V}$	[8]
(c)	CO2	Design a static CMOS logic circuit that implements the logic function $Y = (A+B)C+D$. (indicate both PMOS and NMOS networks)	[4+4]



a) operating mode of Load NMOS,

$$v_{GSL} = v_{DSL}$$

$$\text{as } v_{DSL} > v_{DSL} - V_{TNL}$$

① saturation mode [4 marks]

b) $v_I = 0.2\text{ V}$ case I

$$\text{As } v_I < V_{TND} \rightarrow I_D = 0$$

$$I_D = K_{NL}(v_{DSL} - V_{TNL})^2 = 0$$

$$\Rightarrow v_{DD} - v_o - V_{TNL} = 0$$

$$\Rightarrow v_o = v_{DD} - V_{TNL} = (3 - 0.5)\text{ V} = 2.5\text{ V}$$

① $v_o = 2.5\text{ V}$ [2 marks]

②, ① $I_{DD} = I_{DL} = 0$ [1+1 mark]

$$V_I = 1V \quad \text{Case II}$$

$$I_{DL} = I_{DD} \quad (\text{Assume driver in saturation})$$

$$\Rightarrow (V_{DD} - V_o - V_{TNL})^2 = 8 (V_i - V_{TND})^2$$

$$\Rightarrow (2.6 - V_o)^2 = 8 (1 - 0.4)^2$$

$$\Rightarrow (2.6 - V_o)^2 = 2.88$$

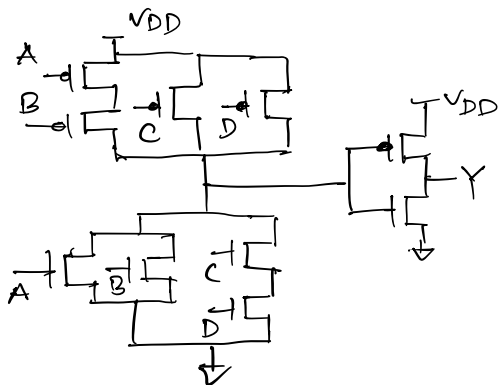
$$\Rightarrow V_o = 2.6 \pm 1.69715 V$$

$$\Rightarrow V_o = \begin{cases} 4.2971 & (\text{Not applicable} > V_{DD}) \\ 0.9029V & > 0.6 \text{ (sat proved)} \end{cases}$$

$$V_o = 0.9029V \quad [2 \text{ marks}]$$

$$I_{DL} = I_{DD} = 0.288 \text{ mA} \quad [1+1 \text{ mark}]$$

$$d) Y = (A+B)+CD \quad [4+4 \text{ marks}]$$



$$V_I = 0.9V \quad \text{Case II}$$

$$I_{DL} = I_{DD} \quad (\text{Assume driver in saturation})$$

$$\Rightarrow (V_{DD} - V_o - V_{TNL})^2 = 8 (V_i - V_{TND})^2$$

$$\Rightarrow (2.5 - V_o)^2 = 8 (0.9 - 0.5)^2$$

$$\Rightarrow (2.5 - V_o)^2 = 3.2$$

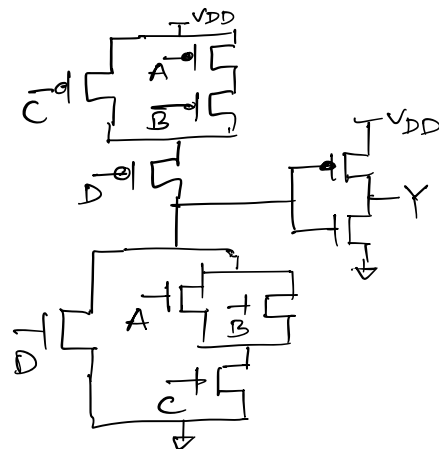
$$\Rightarrow V_o = 2.5 \pm 1.78885 V$$

$$\Rightarrow V_o = \begin{cases} 4.288 & (\text{Not applicable} > V_{DD}) \\ 0.711 & > 0.4 \text{ (sat proved)} \end{cases}$$

$$V_o = 0.711 V \quad [2 \text{ marks}]$$

$$I_{DL} = I_{DD} = 0.1152 \text{ mA} \quad [1+1 \text{ mark}]$$

$$d) Y = (A+B)C + D \quad [4+4 \text{ marks}]$$



Question No. 4
Assume, $V_{DD} = 2.5\text{ V}$, $K_n = 0.2\text{ mA/V}^2$, $K_p = 0.05\text{ mA/V}^2$, $V_{TN} = 0.6\text{ V}$, $V_{TP} = -1\text{ V}$.

(a)	CO2	Identify the operating modes of PMOS and NMOS in figure 4 at point A of the transfer characteristics curve. Then, write down the drain current (I_D) of PMOS and NMOS in terms of V_I , V_O and MOSFET parameters mentioned above.	[8]
(b)	CO2	Calculate the value of V_{OH} and V_{IL} in volts.	[12]

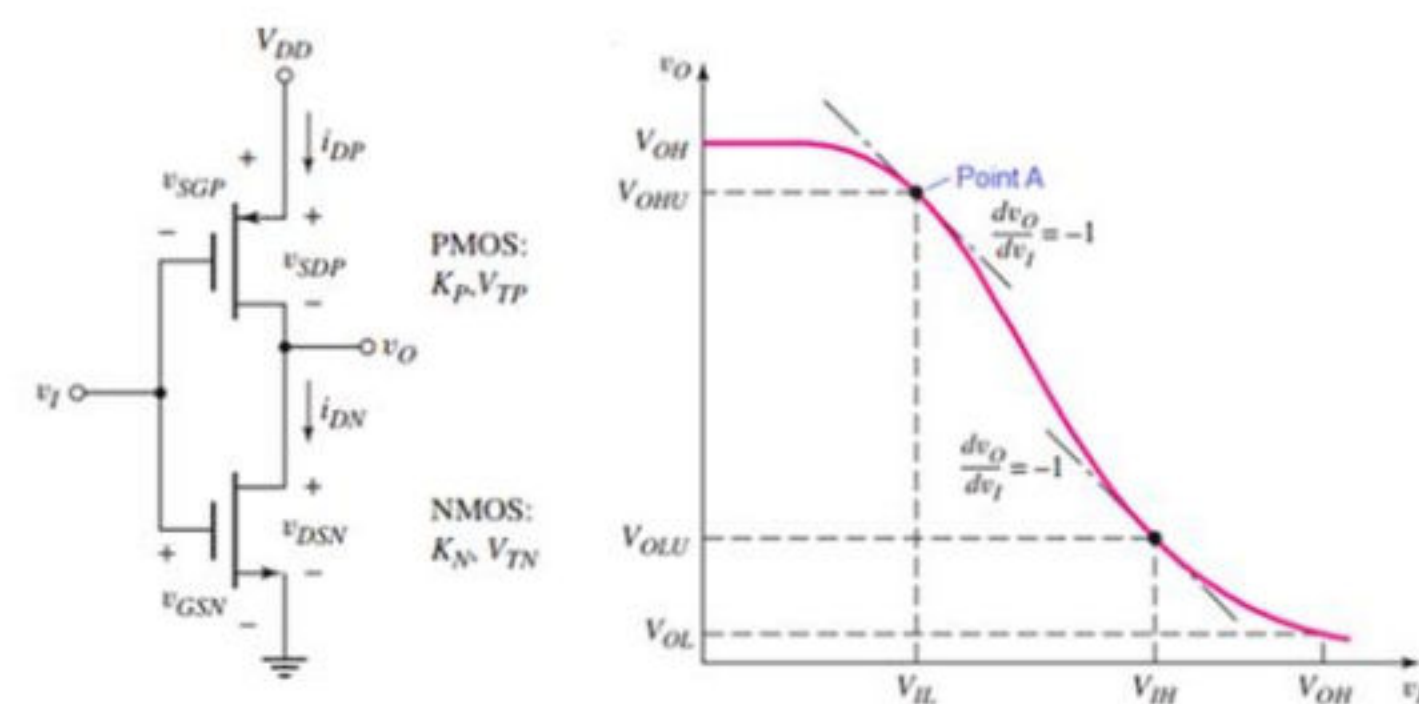


Figure 4

(a) at point A, pmos $\rightarrow$ triode
nmos $\rightarrow$ saturation

$$I_{Dn} = K_n (V_{GS,n} - V_{TN})^2$$

$$= 0.2 (V_I - 0.6)^2$$

$$I_{Dp} = K_p [2 (V_{SG,p} + V_{TP}) V_{SD} - V_{SD}^2]$$

$$= 0.05 [2 (2.5 - V_I - 1) (2.5 - V_O) - (2.5 - V_O)^2]$$

$$= 0.05 [2 (1.5 - V_I) (2.5 - V_O) - (2.5 - V_O)^2]$$

(b) Now $I_{Dp} = I_{Dn}$

$$\Rightarrow 0.2 (V_I - 0.6)^2 = 0.05 [2 (1.5 - V_I) (2.5 - V_O) - (2.5 - V_O)^2]$$

$$\Rightarrow 4 (V_I - 0.6)^2 = [2 (1.5 - V_I) (2.5 - V_O) - (2.5 - V_O)^2] \quad (1)$$

$$\text{Now } \frac{d}{dV_I} [4 (V_I - 0.6)^2] = \frac{d}{dV_I} [2 (1.5 - V_I) (2.5 - V_O) - (2.5 - V_O)^2]$$

$$\Rightarrow 8V_I - 4.8 = 2 (1.5 - V_I) (-1) + 2 (2.5 - V_O) (-1) - 2 (2.5 - V_O) (+1)$$

$$\Rightarrow 8V_I - 4.8 = 7 - 2V_I + 4V_O$$

$$\Rightarrow 10V_I = 4V_O + 2.2$$

$$\Rightarrow V_O = 2.5V_I + 0.55 \quad (ii)$$

Plugging this into eqn (1)

$$4 (V_I - 0.6)^2 = [2 (1.5 - V_I) (2.5 - V_O) - (2.5 - V_O)^2]$$

$$\Rightarrow 4 (V_I - 0.6)^2 = 2 (1.5 - V_I) (2.5 - 2.5V_I - 0.55) - (2.5 - 2.5V_I - 0.55)^2$$

$$\Rightarrow 4V_I^2 - 4.8V_I + 1.44 = 5.85 - 11.4V_I + 5V_I^2 - 3.8025 + 7.75V_I - 6.25V_I^2$$

$$\Rightarrow 5.25V_I^2 - 3.15V_I - 0.6075 = 0$$

$$\therefore V_I = \frac{-0.154 \pm \sqrt{0.7536}}{1.05}$$

$$V_O = 2.5V_I + 0.55$$

$$= 2.5 \times 0.7536 + 0.55$$

$$= 2.434\text{ V}$$

So,

$$V_{IL} = 0.7536\text{ V}$$

$$V_{OH} = 2.434\text{ V}$$

alternate solution:

Final question 1

(a) PMOS: Triode

NMOS: Saturation

$$I_{Dn} = K_n (V_{GS,n} - V_{TN})^2$$

$$= K_n (V_I - V_{TN})^2$$

$$I_{Dp} = K_p [2 (V_{SG,p} + V_{TP}) V_{SD} - V_{SD}^2]$$

$$= K_p [2 (V_{DD} - V_I + V_{TP}) (V_{DD} - V_O) - (V_{DD} - V_O)^2]$$

(b) $I_{Dn} = I_{Dp}$

$$\Rightarrow 0.2 (V_I - 0.6)^2 = 0.05 [2 (2.5 - V_I - 1) (2.5 - V_O) - (2.5 - V_O)^2]$$

$\Rightarrow$ Assume, $V_I - 0.6 = X$

$$2.5 - V_O = Y$$

Then,

$$4X^2 = 2 (0.9 - X) Y - Y^2$$

$$\Rightarrow 4X^2 = 1.8Y - 2XY - Y^2$$

$$\Rightarrow 4X^2 + Y^2 + 2XY - 1.8Y = 0$$

$$4X^2 + Y^2 + 2XY - 1.8Y = 0 \quad (1)$$

$$\frac{dY}{dX} = - \frac{8X + 2Y}{2Y + 2X - 1.8}$$

$$\text{Now, } \frac{dV_O}{dV_I} = \frac{dV_O}{dY} \cdot \frac{dY}{dX} \cdot \frac{dX}{dV_I}$$

$$= (-1) \cdot \frac{dY}{dX} \cdot (1)$$

$$\therefore \frac{dV_O}{dV_I} = - \frac{dY}{dX}$$

$$\text{So, } \frac{dY}{dX} = 1 \Rightarrow -4X - Y = Y + X - 0.9$$

$$\Rightarrow 5X + 2Y = 0.9 \quad (2)$$

Solving (1) & (2) $\Rightarrow$

$$4 \left(\frac{0.9 - 2Y}{5} \right)^2 + Y^2 + 2 \left(\frac{0.9 - 2Y}{5} \right) Y - 1.8Y = 0$$

$$\Rightarrow 4 (0.9 - 2Y)^2 + 25Y^2 + 10 (0.9 - 2Y) Y - 45Y = 0$$

$$\Rightarrow (400 + 25 - 20) Y^2 + (-4 \times 0.9 \times 100 + 9 - 1.8) Y + 100 \times 0.9^2 = 0$$

$\therefore$

$$\Rightarrow (16 + 25 - 20) Y^2 + (-4 \times 0.9 \times 4 + 9 - 45) Y + 4 \times 0.9^2 = 0$$

$$Y = 2.334\text{ V} \rightarrow X = -0.7536\text{ V}$$

$$\text{and } 0.066\text{ V} \rightarrow X = 0.1536\text{ V}$$

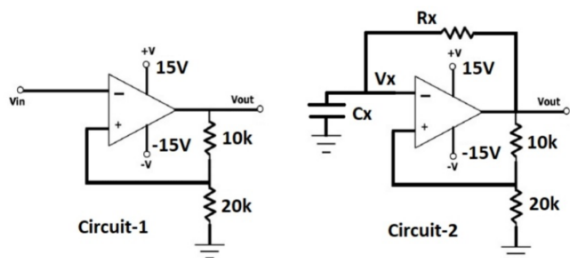
$$V_I = (-V_e)$$

$$V_O =$$

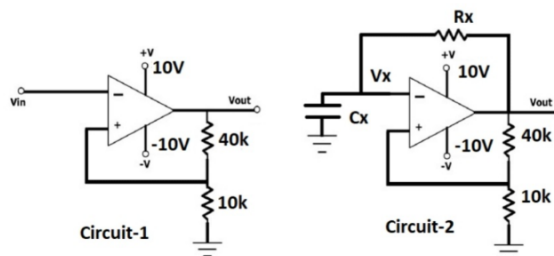
$$V_I = 0.7536\text{ V}$$

$$V_O = 2.434\text{ V}$$

(a)	CO5	Determine the higher threshold voltage V_{TH} and lower threshold voltage V_{TL} of the given Schmitt. Finally determine the hysteresis width of this circuit.	[3+3+2]
(b)	CO5	Given $R_x = 100k\Omega$ and $C_x = 0.621\mu F$, find the duration of the positive half cycle T_1 , and the negative half cycle T_2 . From the determined values of T_1 and T_2 , determine the oscillation frequency f and the duty cycle D of the astable multivibrator. Comment on how the oscillation frequency changes when you double the value of R_x .	[3+3+2 +2+2]



(a)	CO5	Determine the higher threshold voltage V_{TH} and lower threshold voltage V_{TL} of the given Schmitt. Finally determine the hysteresis width of this circuit.	[3+3+2]
(b)	CO5	Given $R_x = 50k\Omega$ and $C_x = 2.466\mu F$, find the duration of the positive half cycle T_1 , and the negative half cycle T_2 . From the determined values of T_1 and T_2 , determine the oscillation frequency f and the duty cycle D of the astable multivibrator. Comment on how the oscillation frequency changes when you halve the value of C_x .	[3+3+2 +2+2]



$$\begin{aligned} \textcircled{a} V_{TH} &= V_H \times \frac{20}{20+10} \\ &= 15 \times \frac{2}{3} \\ &= \boxed{10V} \end{aligned}$$

3 marks

$$\begin{aligned} V_{TL} &= V_L \times \frac{20}{20+10} \\ &= -15 \times \frac{2}{3} \\ &= \boxed{-10V} \end{aligned}$$

3 marks

$$\begin{aligned} \text{Hysteresis width} &= V_{TH} - V_{TL} \\ &= \boxed{20V} \end{aligned}$$

2 marks

$$\begin{aligned} \textcircled{a} V_{TH} &= V_H \times \frac{10}{10+40} \\ &= 10 \times \frac{1}{5} \\ &= \boxed{2V} \end{aligned}$$

3 marks

$$\begin{aligned} V_{TL} &= V_L \times \frac{10}{10+40} \\ &= -10 \times \frac{1}{5} \\ &= \boxed{-2V} \end{aligned}$$

3 marks

$$\begin{aligned} \text{Hysteresis width} &= V_{TH} - V_{TL} \\ &= \boxed{4V} \end{aligned}$$

2 marks

$$\begin{aligned} \textcircled{b} T_1 &= R_x C_x \ln \left(\frac{V_H - V_{TL}}{V_H - V_{TH}} \right) \\ \textcolor{red}{3} &= 100k\Omega \times 0.621\mu F \\ &\quad \times \ln \left(\frac{15+10}{15-10} \right) = \boxed{0.1s} \end{aligned}$$

$$T_2 = R_x C_x \ln \left(\frac{V_L - V_{TH}}{V_L - V_{TL}} \right)$$

$$\begin{aligned} \textcolor{red}{3} &= 100k\Omega \times 0.621\mu F \\ &\quad \times \ln \left(\frac{-15-10}{-15+10} \right) = \boxed{0.1s} \end{aligned}$$

$$\text{Total duration } T = T_1 + T_2 = 0.2s$$

$$\textcolor{red}{2} \text{ Oscillation frequency } = \frac{1}{T} = \boxed{5Hz}$$

$$\textcolor{red}{2} \text{ Duty cycle } = \frac{T_1}{T_1 + T_2} = \boxed{0.5 = 50\%}$$

$\textcolor{red}{2}$ Frequency halves/decreases

$$\begin{aligned} \textcircled{b} T_1 &= R_x C_x \ln \left(\frac{V_H - V_{TL}}{V_H - V_{TH}} \right) \\ \textcolor{red}{3} &= 50k\Omega \times 2.466\mu F \\ &\quad \times \ln \left(\frac{10+2}{10-2} \right) = \boxed{0.05s} \end{aligned}$$

$$T_2 = R_x C_x \ln \left(\frac{V_L - V_{TH}}{V_L - V_{TL}} \right)$$

$$\begin{aligned} \textcolor{red}{3} &= 50k\Omega \times 2.466\mu F \\ &\quad \times \ln \left(\frac{-10-2}{-10+2} \right) = \boxed{0.05s} \end{aligned}$$

$$\text{Total duration } T = T_1 + T_2 = 0.1s$$

$$\textcolor{red}{2} \text{ Oscillation frequency } = \frac{1}{T} = \boxed{10Hz}$$

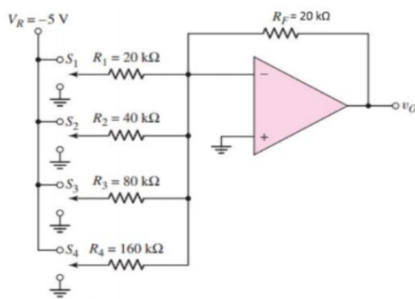
$$\textcolor{red}{2} \text{ Duty cycle } = \frac{T_1}{T_1 + T_2} = \boxed{0.5 = 50\%}$$

$\textcolor{red}{2}$ Frequency doubles/increases

SET-A Q6

Question No. 6

(a)	CO3	Design a 2-bit flash A/D converter circuit to convert an analog signal in the range of 0 V to 5 V. Calculate total number of resistors and comparators required to fabricate this converter and draw the circuit. [Hint: the quantization levels are 0.625V, 1.875V, 3.125V and 4.375V]	[4+6]
(b)	CO3	If at any time instance the input voltage is 3.54 V, find out the 2-bit binary output obtained from the converter circuit designed in (a).	[4]
(c)	CO3	Determine the output voltage, v_o of the 4-bit weighted-resistor D/A in the figure 6 for input = 0111. [Hint: $v_o = V_R \sum \frac{b_n}{2^{n-1}}$]	[6]



① $N = 2 \Rightarrow \text{Levels} = 2^N = 4$

$\therefore$ No. of resistors = 4 (2 marks)

" " comparators = 4 (2 marks)

$\rightarrow \Delta = \frac{5-0}{4} = 1.25V$ (or 3)

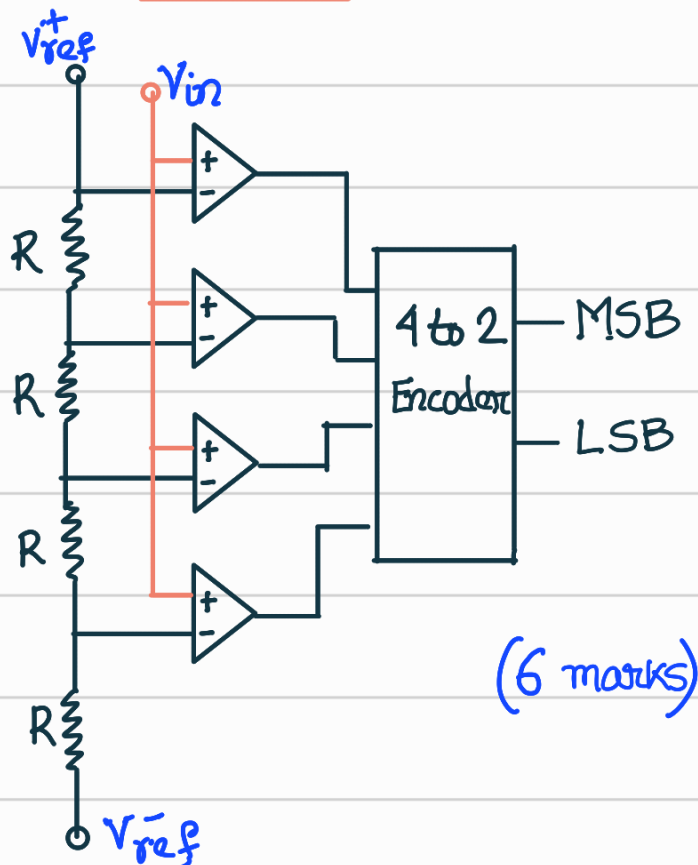
Levels: 0 - 1.25 $\rightarrow$ 0.625V

1.25 - 2.5 $\rightarrow$ 1.875V

2.5 - 3.75 $\rightarrow$ 3.125V

3.75 - 5.0 $\rightarrow$ 4.375V

$V_{ref}^+ = 4.375V$, $V_{ref}^- = -0.625V$



② $V_{in} = 3.54V$

Comparators o/p $\rightarrow$ Low, High, High, High
(0111)

Encoder o/p $\rightarrow$ 10 (4 marks)

③ $b_1 b_2 b_3 b_4$ $V_R = 5V$
 $\downarrow \downarrow \downarrow \downarrow$
0 1 1 1

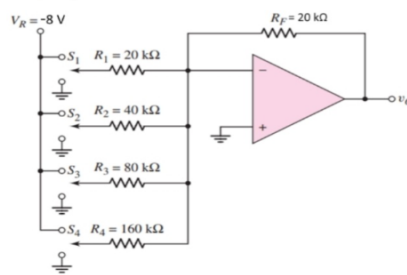
$\therefore V_o = 5 \times \left(0 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8}\right)$

(6 marks) = 4.375V

SET-B Q3

Question No. 3

(a)	CO3	Design a 2-bit flash A/D converter circuit to convert an analog signal in the range of 0 V to 8 V. Calculate total number of resistors and comparators required to fabricate this converter and draw the circuit. [Hint: the quantization levels are 1 V, 3 V, 5 V and 7 V]	[4+6]
(b)	CO3	If at any time instance the input voltage is 4.52 V, find out the 2-bit binary output obtained from the converter circuit designed in (a).	[4]
(c)	CO3	Determine the output voltage, v_o of the 4-bit weighted-resistor D/A in the figure 6 for input = 1011. [Hint: $v_o = V_R \sum \frac{b_n}{2^{n-1}}$]	[6]



① $N = 2 \Rightarrow \text{Levels} = 2^N = 4$

$\therefore$ No. of resistors = 4 (2 marks)

" " comparators = 4 (2 marks)

$\rightarrow \Delta = \frac{8-0}{4} = 2V$ (or 3)

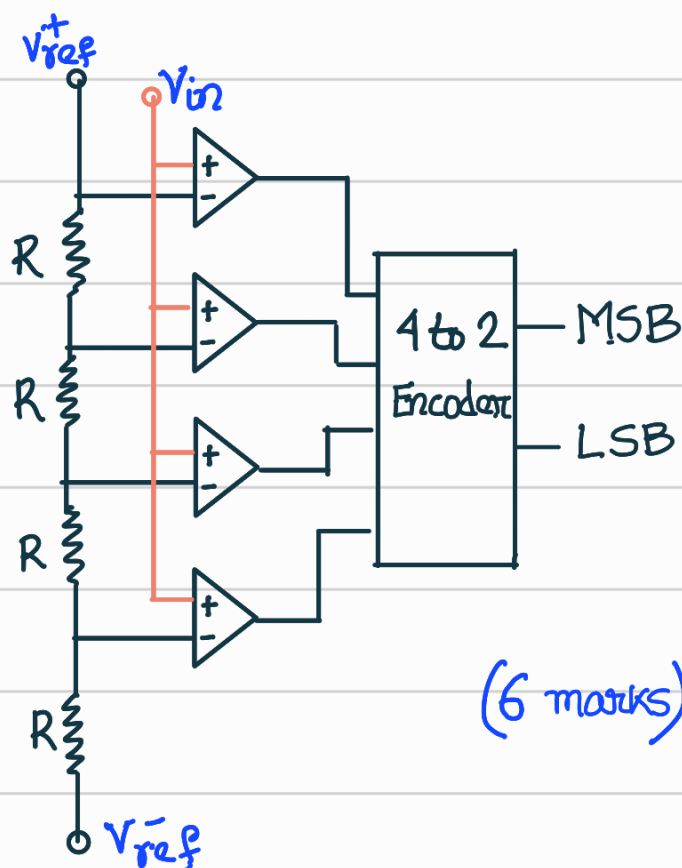
Levels: 0 - 2V $\rightarrow$ 1V

2V - 4V $\rightarrow$ 3V

4V - 6V $\rightarrow$ 5V

6V - 8V $\rightarrow$ 7V

$V_{ref}^+ = 7V$, $V_{ref}^- = -1V$



② $V_{in} = 4.52V$

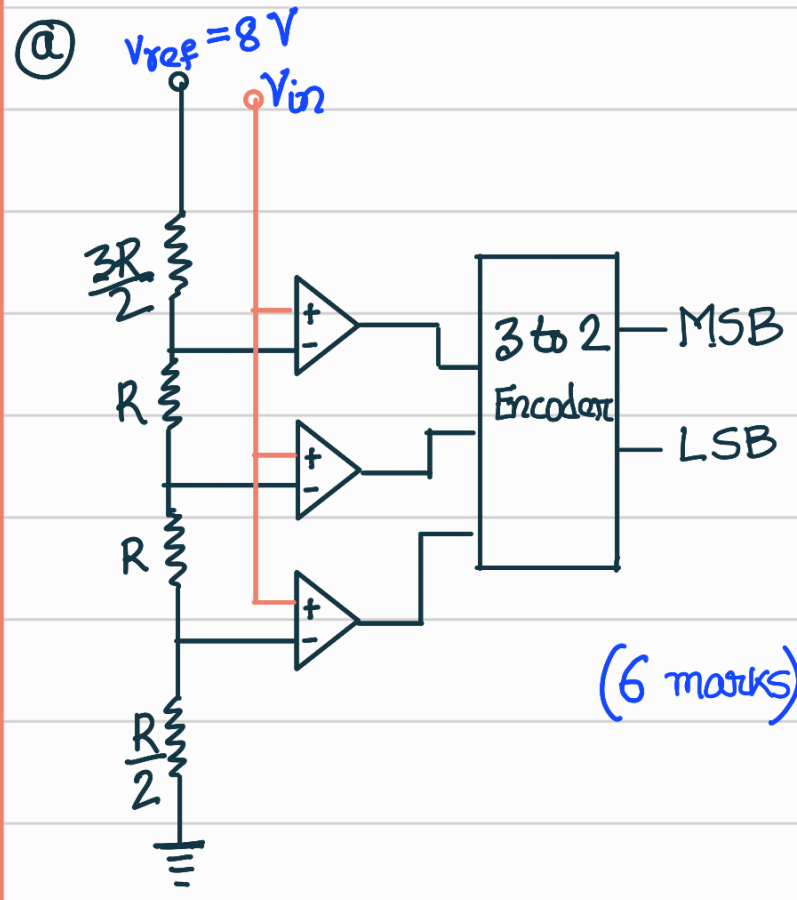
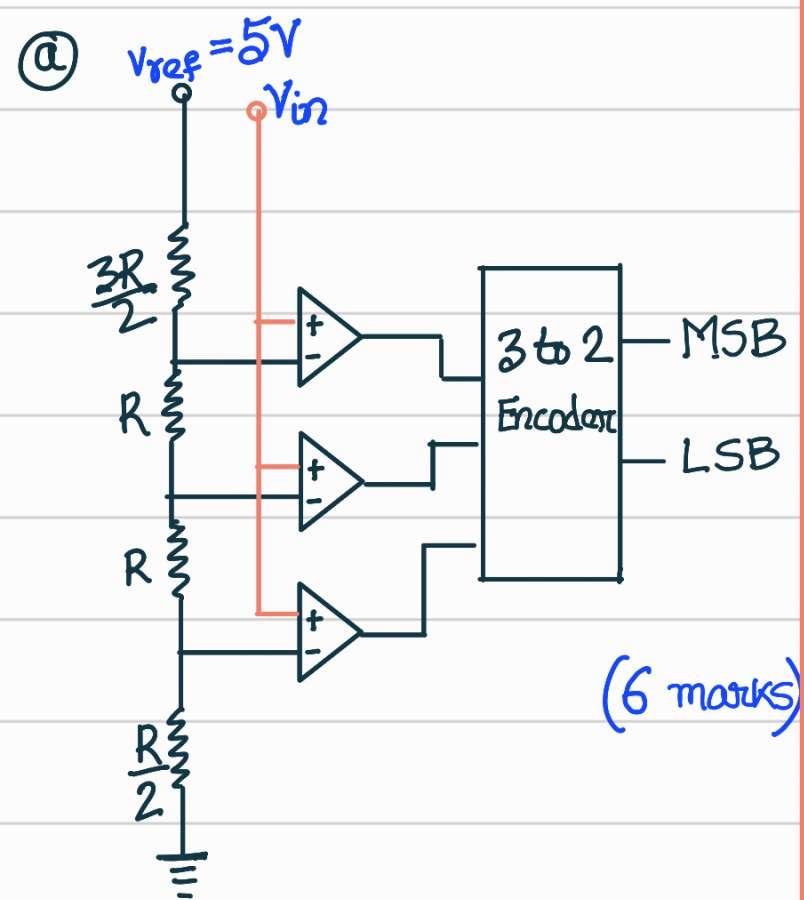
Comparators o/p $\rightarrow$ Low, Low, High, High
(0011)

Encoder o/p $\rightarrow$ 01 (4 marks)

③ $b_1 b_2 b_3 b_4$ $V_R = 8V$
 $\downarrow \downarrow \downarrow \downarrow$
1 0 1 1

$\therefore V_o = 8 \times \left(1 + 0 + \frac{1}{4} + \frac{1}{8}\right)$

= 11V (6 marks)



b) $V_{in} = 3.54V$
 Comparators o/p $\rightarrow$ High, High, High
 (111)
 Encoder o/p $\rightarrow$ 11 (4 marks)

b) $V_{in} = 4.52V$
 Comparators o/p $\rightarrow$ Low, High, High
 (011)
 Encoder o/p $\rightarrow$ 10 (4 marks)